

Ningchen MA

930 Seymour St, Vancouver, BC, Canada, V6B 1B4
(+1) - 438 - 530 - 7810 | ningchen.ma@yale.edu

EDUCATION

Yale University

M.S. in *Computer Science*

New Haven, CT

Sep.2026 – May.2027

McGill University

B.S. in *Computer Science (Honours)*

Montreal, Canada

Sep.2017-Jun.2022

- **First Class Honours** in Computer Science | **GPA: 3.87/4.00**

WORK EXPERIENCE

Amazon

Software Development Engineer (SDE), Dept. of PXT

Vancouver, Canada

Oct.2022 – Jun.2026

- **Led backend development for a large-scale production workforce planning and analytics system**, owning API design, data modeling, business logic, testing, deployment, and operational support.
 - Designed and implemented 40+ GraphQL APIs supporting data setup, scenario planning, workforce metric calculations, and complex planning workflows.
 - Built core modeling logic for hiring, attrition, promotions, and organizational changes, ensuring accurate, conflict-free headcount projections across multiple planning dimensions.
 - Designed scalable DynamoDB data models and backend services to support multi-dimensional organizational data, improving performance, consistency, and reliability.
- **Developed a React-based frontend for an internal service**, implementing user-facing features and integrating with RESTful APIs.
- **Developed a GenAI-powered analytics assistant**, implementing both the AI agent and its supporting remote MCP server end-to-end, enabling business users to query workforce insights via natural language and significantly reducing manual data exploration time from tens of minutes to seconds

RESEARCH EXPERIENCE

Interest: *Computer Architecture, Networks, Software System Design*

Shir Group, McGill University

Research Assistant, Directed by Dr. Jonathan Van der Cruysse & Prof. [Christophe Dubach](#)

May.2022 – Sep.2022

- Optimized compiler code generation using equality saturation to enhance system performance and code efficiency.
- Developed heuristics to select the most effective Intermediate Representation (IR) for optimal Neural Processing Unit (NPU) utilization.
- Created a Swift code generation toolchain utilizing the MLCompute library to boost execution performance on Apple's M-series neural accelerator chip.

Anrl Cloud Computing/IoT Group, McGill University

Research Assistant, Supervised by Prof. [Muthucumaru Maheswaran](#)

Jan.2022 – Apr.2022

- Configured ESP32 micro-controllers with RIOT, a real-time mini OS, optimizing for MQTT communications in edge computing scenarios.
- Streamlined communications processes between devices, enhancing real-time data handling and system efficiency.

Teaching Assistant, Dept. of Computer Science

Course: *Functional Programming (COMP 302)*

Jan.2022-Apr.2022

- Provided weekly office hours to help over 50 students in course review, deeply understand course materials and solve problems.
- Assisted the teacher in assignment design and graded students' assignments and exams.

PROJECT EXPERIENCE

A Mini Compiler

Independent Designer, Developer

Oct.2018 – Jan.2019

- Programmed in Java to develop a lexer to tokenize C-like source code, breaking it into meaningful components for further analysis.
- Implemented a parser to ensure the grammatical correctness of the tokens produced by the lexer, generating an Abstract Syntax Tree (AST) that represents the input program's structure.
- Engineered a code generator to translate the AST into assembly code by utilizing the Chaitin algorithm, converting high-level C-like constructs into executable machine instructions.
- Optimized the compiler to eliminate dead code and enhance the register allocation process, improving overall performance and efficiency.

Design and Implement of a Basic Operating System in C

Independent Designer, Developer

Oct.2018 – Jan.2019

- Developed a UNIX-style operating system with a command-line user interface.
- Realized a CPU scheduler using a round-robin algorithm to efficiently manage CPU time frames among multiple tasks.
- Designed a Process Control Block (PCB) to store task context and manage task states.
- Engineered memory management using an array to handle read and write operations driven by tasks.

SKILLS

Programming: Java, Python, Scala, C/C++, OCaml, MIPS Assembly, Bash Script, HTML+CSS, JavaScript, PHP

Web / APIs: React, JavaScript, GraphQL, REST APIs

Amazon Web Services: Lambda, DynamoDB, API Gateway, OpenSearch, ECS, Route 53, VPC